

AeroForensics AI Stack Technology

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AeroForensics AI — Technology Stack & Architecture

Project Overview

AeroForensics AI is a real-time environmental enforcement command center dashboard. It integrates live sensor telemetry, AI-powered anomaly detection, forensic evidence management, vehicle emission intelligence, pollution source attribution, and multi-language citizen-facing e-Challan services.

Frontend Technologies

	Technology	Version	Purpose	
	React	18.x	UI framework	
	TypeScript	~5.9.3	Type-safe development	
	Vite (rolldown-vite)	7.3.1	Build tool & dev server	
	Tailwind CSS	3.4.11	Utility-first CSS	
	shadcn/ui	latest	Component library (Radix UI + Tailwind)	
	Radix UI	various	Headless accessible UI primitives	
	React Router DOM	7.9.5	Client-side routing	
	Recharts	2.15.4	Data visualization (line/bar charts)	
	Sonner	2.0.7	Toast notifications	
	Lucide React	0.576.0	Icon library	
	Framer Motion (via motion)	12.23.25	Animation library	
	React Hook Form	7.66.0	Form state management	
	Zod	3.25.76	Schema validation	
	date-fns	3.6.0	Date formatting & manipulation	
	axios	1.13.1	HTTP client	
	ky	1.13.0	Modern HTTP client	
	React Helmet Async	2.0.5	Document head management	
	React Dropzone	14.3.8	File drag-and-drop	
	QRCode	1.5.4	QR code generation	
	Embla Carousel	8.6.0	Carousel/slider component	

	Sentry React	9.47.1	Error monitoring & tracking	
	Miaoda React DevKit	0.1.1-beta.14	Platform integration SDK	
	Miaoda SC Plugin	1.0.62	Platform service client	

Dev Tools

	Tool	Purpose	
	Biome	Linting & formatting	
	TypeScript (ts go)	Type checking	
	ast-grep	Custom rule checking	
	PostCSS	CSS processing	
	Autoprefixer	CSS vendor prefixes	
	Vite Plugin SVGR	SVG as React components	
	Tailwind Container Queries	CSS container query support	

Backend Technologies

	Technology	Version	Purpose	
	Supabase	2.x	Backend-as-a-Service (BaaS)	
	PostgreSQL	15+	Primary database	
	Supabase Edge Functions	Deno 2.x	Serverless TypeScript functions	
	Supabase Realtime	—	Live WebSocket subscriptions	
	Supabase Storage	—	File/blob storage	
	Supabase Auth	—	Authentication & authorization	
	Row Level Security (RLS)	—	Database-level access control	

Edge Functions Deployed

	Function	Purpose	Tech	
	anomaly-detecti on	Automated sensor threshold scanning & anomaly insertion	Deno + Supabase JS client	

	fastag-service	NPCI FASTag balance checks, fine deduction, UPI linking	Deno + mock NPCI DB	
	ml-attribution	Random Forest pollution source classifier	Deno + hand-coded ML (5-tree RF)	

Database Schema (Supabase PostgreSQL)

Tables

- sensors** — Sensor metadata (location, sector, status)
- sensor_readings** — Live telemetry (SO₂, PM2.5, NOx, temperature, wind)
- anomalies** — Detected environmental anomalies with severity & status
- evidence_packets** — Generated forensic evidence case files
- translations** — Multi-language UI string dictionary (en/hi/kn)
- user_preferences** — Per-user language & alert threshold settings
- incident_rooms** — Collaboration room metadata
- room_messages** — Chat messages per room
- room_participants** — Room membership & presence

Storage Buckets

- evidence-exports** — PDF/JSON evidence packet files
- public** — Public assets

APIs & External Services

Supabase APIs (Frontend → Backend)

	API	Usage	
	<code>supabase.from('sensor_readings').select(...)</code>	Live telemetry queries	
	<code>supabase.from('anomalies').select(...)</code>	Anomaly data fetch	
	<code>supabase.from('translations').select(...)</code>	i18n dictionary loading	
	<code>supabase.functions.invoke('anomaly-detection')</code>	Trigger anomaly scan	
	<code>supabase.functions.invoke('fastag-service')</code>	FASTag balance & payments	

	<code>supabase.functions.invoke('ml-attribution')</code>	ML pollution source inference	
	<code>supabase.storage.from('evidence-exports').upload(...)</code>	Evidence file upload	
	<code>supabase.storage.from('evidence-exports').download(...)</code>	Evidence file download	
	<code>supabase.channel('postgres_changes', ...).subscribe()</code>	Real-time data streams	

External APIs

	Service	Integration	Purpose	
	Google Maps Embed API	iframe embed	Map visualization (place mode)	
	NPCI FASTag (simulated)	Edge Function mock	Vehicle toll balance & fine deduction	
	Sentry	Error tracking SDK	Production error monitoring	

AI / ML Technologies

Machine Learning Model

	Model	Type	Details	
	Random Forest Classifier	Hand-coded TypeScript	5 decision trees, 4 features	
	Accuracy	87%	On 2,400 synthetic training samples	
	Version	rf-v1.2.0	Deployed in <code>ml-attribution</code> Edge Function	
	Features	PM2.5 delta, wind corridor alignment, traffic density, industrial proximity, construction activity, wind speed		
	Outputs	Traffic (0), Construction (1), Industrial (2) — softmax probabilities		
	Feature Importance	Tracked per-tree	Returned in API response	

AI / LLM Integrations

	LLM / AI Service	Integration	Purpose	
	Gemini 2.5 Flash	Miaoda LLM skill (large-language-model)	Content generation, text understanding	
	AI Search (Gemini + Google Grounding)	Miaoda skill (ai-search)	Web-aware intelligent search with citations	
	Image Generation	Miaoda skill (image-generation)	AI image creation from text prompts	
	OCR (OCR.space)	Miaoda skill (ocr-space)	Text extraction from images/PDFs	
	Text-to-Speech (LemonFox)	Miaoda skill (text-to-speech)	Audio narration generation	
	Speech-to-Text (Whisper v3)	Miaoda skill (speech-to-text)	Audio transcription	

Note: LLM integrations are available via the Miaoda platform skill system. They are wired through Supabase Edge Functions or direct frontend hooks and can be invoked on demand.

Multi-Language Support (i18n)

	Language	Code	Coverage	
	English	en	Full	
	Hindi	hi	Full (40+ keys)	
	Kannada	kn	Full (40+ keys)	

- Dictionary stored in Supabase **translations** table
- Fallback dictionary in **src/hooks/useI18n.ts**
- Language switcher UI in dashboard header

Application Features

- Live Telemetry Dashboard** — Real-time sensor readings with Supabase subscriptions
- Anomaly Detection** — Automated threshold-based anomaly scanning with toast alerts
- Historical Anomaly Archive** — Searchable, filterable, exportable anomaly history
- Evidence Locker** — Forensic evidence packet generation & Supabase Storage upload
- Dispatch Logistics** — Unit dispatch simulation with ETA tracking
- Notification System** — Bell icon with badge, popover panel, severity filtering, bulk actions
- Collaboration Rooms** — Real-time chat with presence indicators
- Pollution Source Attribution Engine** — ML-powered traffic/industrial/construction classification

- Vehicle Emission Intelligence Engine — ANPR simulation, emission class detection (BS-III/IV/VI), CO₂ estimates, pollution hotspot map
- Industrial AQI Panel — Factory gate monitors with compliance charts
- Citizen e-Challan App — Mobile-optimized challan list, FASTag/UPI payments, vehicle registration
- Multi-Language Support — English, Hindi, Kannada with live switching

Development & Deployment

	Aspect	Details	
	Package Manager	pnpm	
	Node Version	20+	
	Lint Command	<code>npm run lint</code> (tsgo + biome + ast-grep + tailwind check + vite build test)	
	Dev Server	<code>npx vite --host 0.0.0.0 --port 5173</code>	
	Build Output	<code>/workspace/.dist</code>	
	PM2 Process	<code>vite-sandbox</code>	
	Git Remote	<code>https://github.com/niveditha819/AeroForensics-AI</code>	
	CI/CD	Manual push to GitHub	

File Structure

```
aero-forensics-ai/
├── src/
│   ├── pages/
│   │   ├── AeroForensicsPage.tsx    # Main dashboard (~1800 lines)
│   │   └── CitizenAppPage.tsx      # Citizen e-Challan app
│   ├── components/
│   │   ├── PollutionSourceAttribution.tsx  # ML attribution engine
│   │   ├── VehicleEnforcementPanel.tsx      # Vehicle emission intelligence
│   │   ├── IndustrialAQIPanel.tsx          # Factory AQI monitors
│   │   └── ui/                             # shadcn/ui components
│   ├── hooks/
│   │   ├── useI18n.ts                   # i18n hook (en/hi/kn)
│   │   ├── useSensors.ts                # Sensor real-time subscriptions
│   │   ├── useAnomalies.ts              # Anomaly real-time subscriptions
│   │   ├── useRooms.ts                  # Collaboration room hooks
│   │   └── useEvidence.ts                # Evidence export utilities
│   ├── db/
│   │   └── supabase.ts                  # Supabase client init
│   ├── routes.tsx                       # React Router config
│   └── main.tsx                          # App entry point
├── supabase/
│   ├── migrations/                      # Database schema & seed SQL
│   └── functions/                       # Edge Functions (Deno)
│       ├── anomaly-detection/          # Threshold-based anomaly scanner
│       └── fastag-service/              # FASTag balance & payments
```

■	■■■ ml-attribution/	# Random Forest classifier
■■■	public/	# Static assets
■■■	index.html	# HTML entry
■■■	package.json	# Dependencies
■■■	tailwind.config.js	# Tailwind configuration
■■■	README.md	# Project overview

License

Proprietary — Environmental Enforcement AI Dashboard by Miaoda Platform.